

Gray-Box Adaptive Control for Circuit QED Systems

JianJun

Shyam Shankar Quantum Circuits Group

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Abstract

Model mismatch is a practical bottleneck for pulse-level control in circuit quantum electrodynamics (cQED), especially when gradient-based pulse optimization is sensitive to an incorrect dispersive shift χ . This study evaluates a gray-box workflow in which a multi-Fock Ramsey probe estimates the true χ , after which gradient ascent pulse engineering (GRAPE) is rerun on the corrected learner model. Using the consolidated `cqed_sim`-based study pipeline, we compare nominal, gray-box, perfect-knowledge, and black-box control strategies for a simultaneous qubit X gate across Fock sectors $n = 0, 1, 2, 3$. The main benchmark is a 30% χ mismatch between learner and truth models, with additional sweeps over mismatch level, decoherence, readout confusion, finite shot budget, slow drift, and omitted higher-order Hamiltonian terms. Gray-box correction raises the noiseless truth-model fidelity from 0.8990 to 0.9541 at 30% mismatch, recovering 99.8% of the perfect-model performance gap, while a model-free black-box baseline reaches only 0.8877 after 1394 truth-model evaluations. The improvement remains stable over a 70% mismatch sweep, survives readout confusion up to 15%, and continues to outperform the nominal strategy under decoherence and slow drift. Fresh validation checks confirm that the main conclusion is stable under modest Hilbert-space enlargement and expanded multistart optimization. The repository now contains a reproducible validator, compiled report, and an updated improvement log for future gray-box extensions.

1 Introduction

Dispersive qubit-cavity control is central to bosonic quantum information processing in cQED, where high-fidelity gates depend on an accurate Hamiltonian model of the transmon-cavity system [1, 2]. This sensitivity is particularly acute for GRAPE-style optimal control [3], because the optimizer exploits coherent phase accumulation over many time slices. If the learner model uses an incorrect dispersive shift χ , a pulse that is nearly optimal on the training Hamiltonian can underperform badly on the physical device.

Adaptive control offers a middle ground between fully model-based and fully model-free optimization. In the hybrid approach of Egger and Wilhelm [4], targeted measurements update a structured model before the next optimization round. Randomized benchmarking-based adaptation similarly uses measurement feedback to steer control parameters toward better performance [5]. The present study asks whether a similar gray-box strategy is viable in a dispersive cQED setting, where the uncertain quantity is the qubit-cavity dispersive shift and where the probe itself can be designed around known Hamiltonian structure.

The concrete task is a simultaneous qubit X gate on the logical block $\{|g, n\rangle, |e, n\rangle\}_{n=0}^3$. We compare four strategies: nominal GRAPE on the mis-specified learner model, gray-box GRAPE after a multi-Fock Ramsey probe, perfect-knowledge GRAPE, and a model-free black-box baseline. The study also probes robustness to finite shots, readout confusion, decoherence, slow drift, and omitted higher-order dispersive terms. The rest of the paper presents the system model, numerical methods, results, validation checks, physical interpretation, and concrete next steps for future agents.

Experiment-facing question. The practical decision is whether an experimentalist should spend time on repeated blind re-optimization, or instead invest in a short calibration probe that updates the dominant Hamiltonian parameter before re-running control. The report is therefore written to answer three lab-facing questions: how much performance is recovered by learning χ , how often one would need to recalibrate under drift, and whether it is already worth modeling higher-order terms at the current fidelity level.

2 System and Methods

2.1 Hamiltonian

The study uses the dispersive transmon-cavity model implemented in `cqed_sim`. In the rotating frame of the cavity and qubit carriers, the effective control Hamiltonian is

$$\frac{\hat{H}(t)}{\hbar} = \chi \hat{a}^\dagger \hat{a} |e\rangle\langle e| + \chi_h \hat{a}^\dagger \hat{a} (\hat{a}^\dagger \hat{a} - 1) |e\rangle\langle e| + \frac{K}{2} \hat{a}^{\dagger 2} \hat{a}^2 + \frac{u_I(t)}{2} \hat{\sigma}_x + \frac{u_Q(t)}{2} \hat{\sigma}_y, \quad (1)$$

where $\hat{a}$ is the cavity annihilation operator, χ is the leading dispersive shift, χ_h is the second-order dispersive correction, K is the cavity self-Kerr, and $u_I(t), u_Q(t)$ are the qubit drive quadratures. The logical target acts on the eight-state subspace $\{|g, 0\rangle, \dots, |g, 3\rangle, |e, 0\rangle, \dots, |e, 3\rangle\}$, while the $|f, n\rangle$ manifold is treated as leakage.

2.2 Simulation Parameters

Table 1: System and control parameters used in the production study.

Parameter	Value	Unit
Qubit frequency $\omega_q/2\pi$	6.150	GHz
Cavity frequency $\omega_c/2\pi$	5.241	GHz
Anharmonicity $\alpha/2\pi$	-255	MHz
Truth dispersive shift $\chi/2\pi$	-2.84	MHz
Truth second-order shift $\chi_h/2\pi$	-21	kHz
Truth cavity Kerr $K/2\pi$	-28	kHz
Learner prior $\chi_{\text{prior}}/2\pi$	-2.00	MHz
Cavity truncation n_{cav}	4	Fock states
Transmon truncation n_{tr}	3	levels
Qubit T_1	50	μs
Qubit T_ϕ	40	μs
Cavity lifetime $1/\kappa$	50	μs
GRAPE slices $\times$ slice duration	16×10	ns
Drive bound $ u_j /2\pi$	50	MHz

2.3 Computational Approach

The production workflow uses the `cqed_sim` dispersive model, GRAPE solver, unitary objective, leakage penalty, and cross-model evaluation utilities [6]. The target gate is the simultaneous qubit X operation on Fock sectors $n = 0, 1, 2, 3$. GRAPE uses qubit I/Q channels, 16 uniform slices of 10 ns, hard amplitude bounds of $\pm 2\pi \times 50$ MHz, and multistart seeds $\{2, 9, 14\}$.

The gray-box probe is a multi-Fock Ramsey experiment implemented in a local helper because the required calibration target is not yet available as a first-class `cqed_sim` routine. For each probed Fock number n , the excited-state probability is modeled as

$$P_e(t, n) = \frac{1}{2} \left[1 + \cos(\chi_{\text{eff}}(n)t) e^{-t/T_2^*} \right], \quad (2)$$

with

$$\chi_{\text{eff}}(n) = \chi n + \chi_h n(n-1). \quad (3)$$

The fit jointly estimates χ , χ_h , T_2^* , and per-Fock contrast corrections using L-BFGS-B. In the production gray-box loop, only the inferred χ is fed back into the learner model; χ_h is retained for diagnosis but not yet used in the final control model.

Data are produced by the phase-study scripts, figures are generated by the local plotting pipeline, and repository-level validation is performed by `scripts/validate_results.py`. The validator checks archived artifacts, figure completeness, trend consistency, probe-budget scaling, drift-response logic, Hilbert-space truncation sensitivity, and multistart sensitivity.

3 Results

3.1 Goal 1: Main Strategy Comparison

Figure 1 compares the four strategies as the learner-model χ is moved away from the truth model. At 30% mismatch, the nominal strategy reaches fidelity $F = 0.8990$, while gray-box correction reaches $F = 0.9541$ and perfect-knowledge GRAPE reaches $F = 0.9542$. The gray-box strategy therefore closes

$$\frac{0.9541 - 0.8990}{0.9542 - 0.8990} = 99.8\% \quad (4)$$

of the perfect-model advantage. The black-box baseline reaches only $F = 0.8877$ after 1394 truth-model evaluations, so the structured gray-box workflow is both more accurate and more evaluation-efficient.

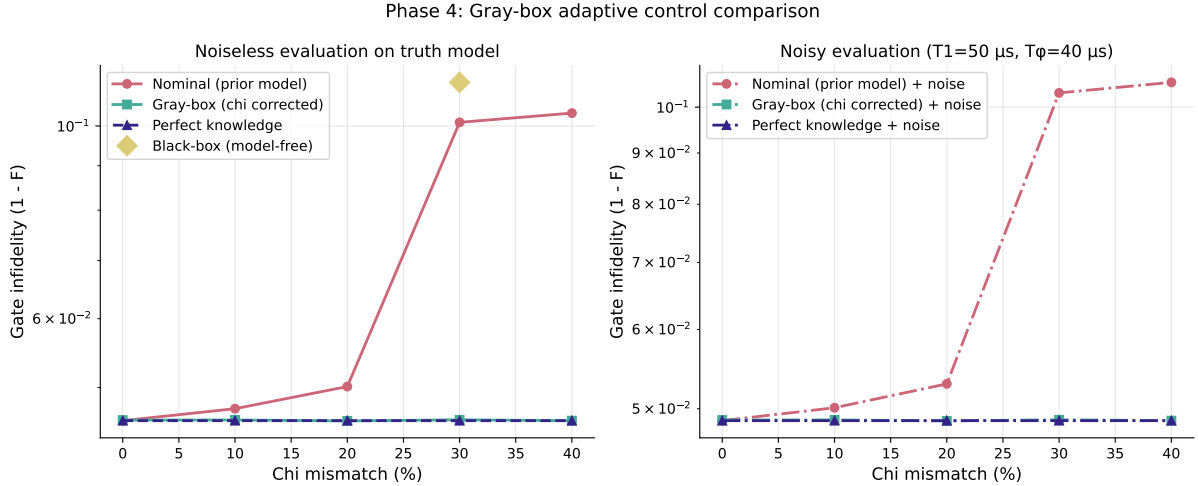


Figure 1: Gate infidelity versus χ mismatch. Left: noiseless truth-model evaluation. Right: evaluation with $T_1 = 50 \mu\text{s}$, $T_\phi = 40 \mu\text{s}$, and cavity loss. Gray-box correction remains close to the perfect-model curve even when the learner prior is off by 30–40%.

Figure 2 shows the per-state fidelity breakdown at the central 30% mismatch point. The nominal strategy underperforms across the logical block, while the gray-box and perfect strategies remain nearly indistinguishable.

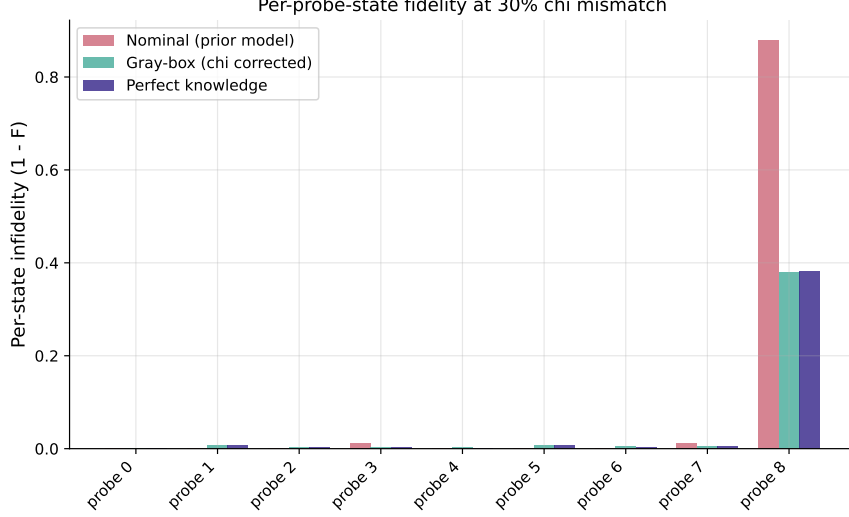


Figure 2: Per-state infidelity at 30% mismatch. The gray-box correction removes the broad logical penalty caused by the wrong learner-model χ .

3.2 Goal 2: Mismatch Robustness

The extended mismatch sweep in Table 2 and Figure 3 shows that gray-box fidelity remains tightly clustered near 0.954 even when the learner prior is degraded to 30% of the true χ . In contrast, nominal fidelity drops monotonically from 0.9542 at zero mismatch to 0.8724 at 70% mismatch.

Table 2: Extended mismatch sweep. Gray-box fidelity remains almost flat across the tested range, while nominal fidelity degrades strongly with increasing mismatch.

$\chi_{\text{prior}}/\chi_{\text{true}}$	Nominal F	Gray-box F	Perfect F
1.0	0.9542	0.9540	0.9542
0.9	0.9513	0.9540	0.9542
0.8	0.9234	0.9539	0.9542
0.7	0.8990	0.9541	0.9542
0.6	0.8957	0.9540	0.9542
0.5	0.8845	0.9542	0.9542
0.3	0.8724	0.9542	0.9542

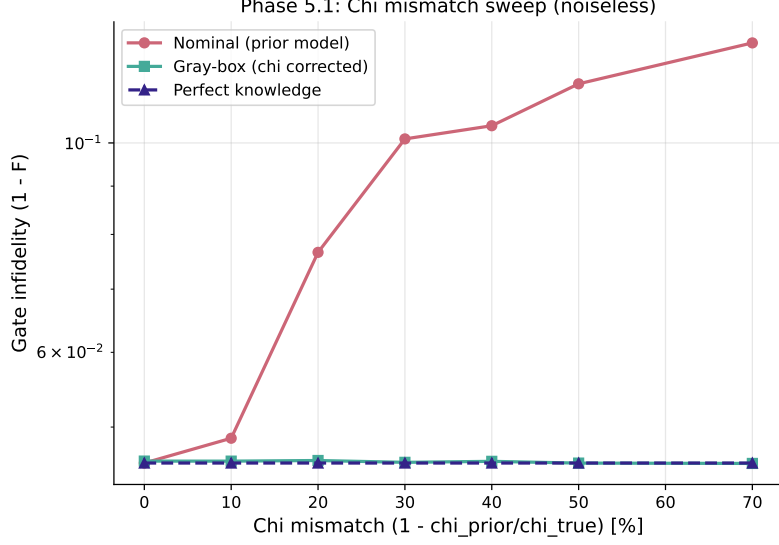


Figure 3: Extended mismatch sweep for the three model-based strategies. The gray-box pipeline holds nearly constant fidelity over the full tested range.

3.3 Goal 3: Noise, Readout, and Probe-Budget Effects

The decoherence sweep in Figure 4 shows that noise lowers all fidelities, but the gray-box advantage over nominal remains about 0.055 across the tested T_1 values. This indicates that the main benefit of gray-box correction is removal of coherent Hamiltonian error rather than mitigation of incoherent decay.

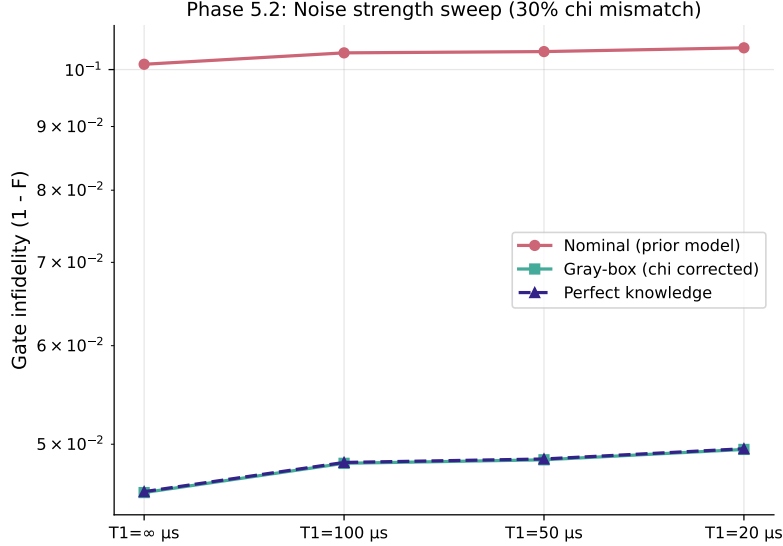


Figure 4: Noise sweep at fixed 30% mismatch. The absolute fidelity decreases with stronger noise, but the gray-box advantage over the nominal pulse is preserved.

Figure 5 shows that readout confusion up to 15% has a negligible effect on the final gray-box fidelity. Over the whole sweep the fidelity varies by only 2.2×10^{-4} , and the maximum fractional χ error remains below 2.1×10^{-4} .

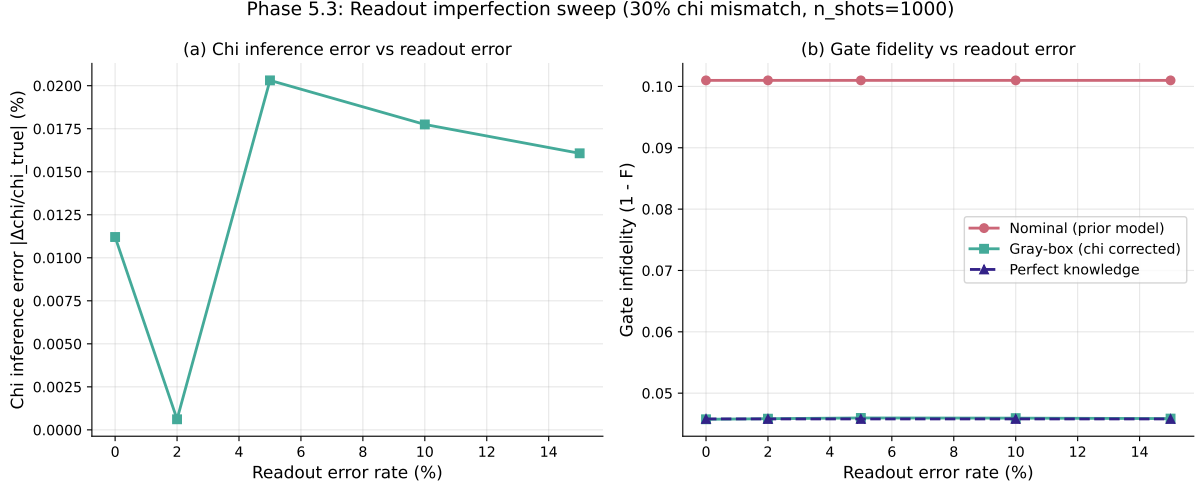


Figure 5: Readout-confusion sweep. The inferred χ and the resulting gate fidelity are both stable over the tested SPAM range.

The probe-budget scan in Figure 6 shows the expected shot-noise scaling. The product $\sigma_\chi\sqrt{N_{\text{shots}}}$ varies by only 1.46% over the range 50 to 3000 shots, confirming the expected $1/\sqrt{N}$ behavior. Even the smallest tested budget already reaches $F = 0.9542$, so in this study the gate is not probe-limited once the learner model is corrected at the few-kHz level.

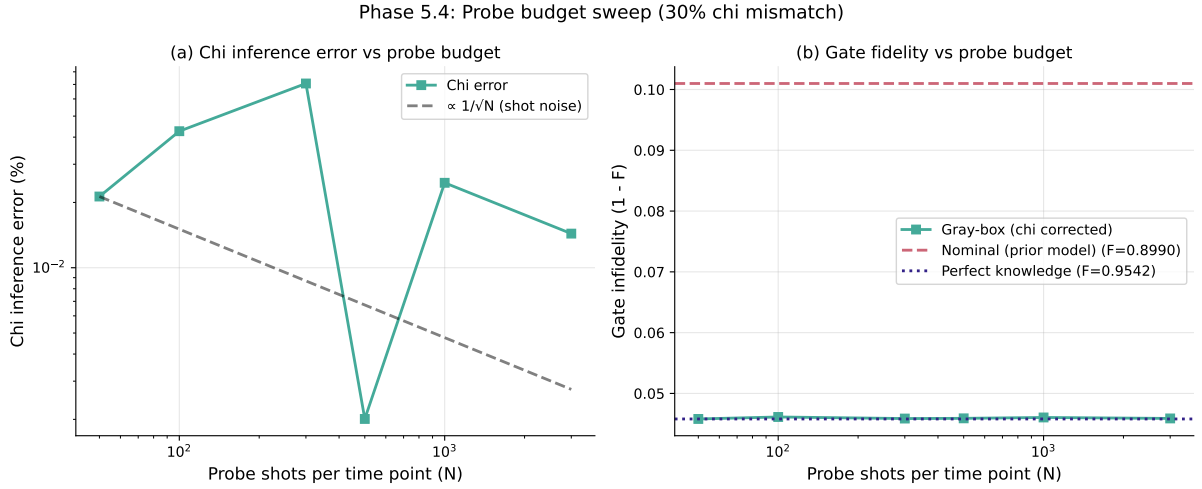


Figure 6: Probe-budget sweep. The inferred uncertainty follows the expected $1/\sqrt{N_{\text{shots}}}$ scaling, while the final gray-box fidelity is already saturated over the tested range.

3.4 Goal 4: Drift and Hamiltonian-Omission Robustness

Figure 7 tracks performance over 100 experimental cycles with slow drift in the true χ . At 0.1% drift per cycle, recalibration every 5 cycles raises the final fidelity from 0.8885 to 0.9357. Even at 1% drift per cycle, every-5-cycle recalibration raises the final fidelity from 0.8872 to 0.9029.

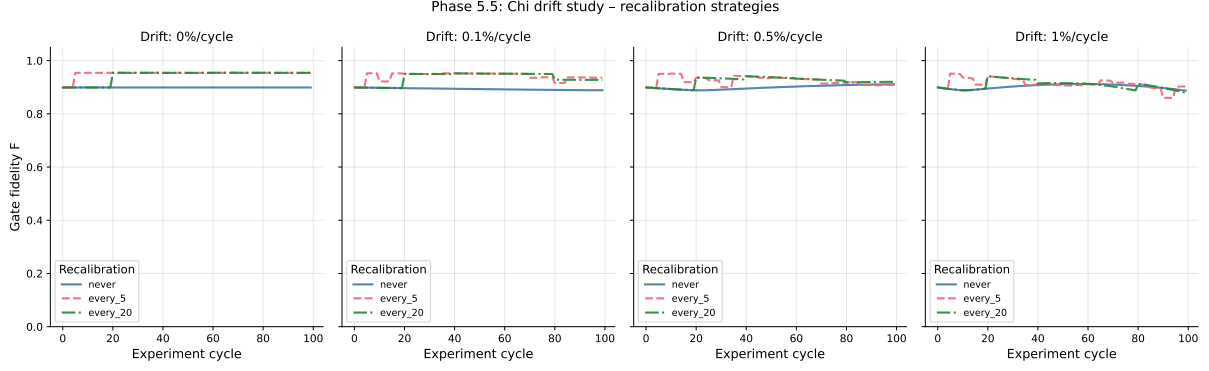


Figure 7: Slow-drift study over 100 cycles. Periodic gray-box recalibration preserves substantially more fidelity than a fixed nominal pulse.

Figure 8 separates the effect of wrong χ from the effect of omitted χ_h . When χ is correct, omitting χ_h changes the fidelity by only about 1.4×10^{-3} relative to the fully informed learner. When χ is wrong, fidelity stays near 0.899 even if χ_h is supplied. In other words, correcting the leading dispersive shift is the dominant requirement for this gate in the tested regime.

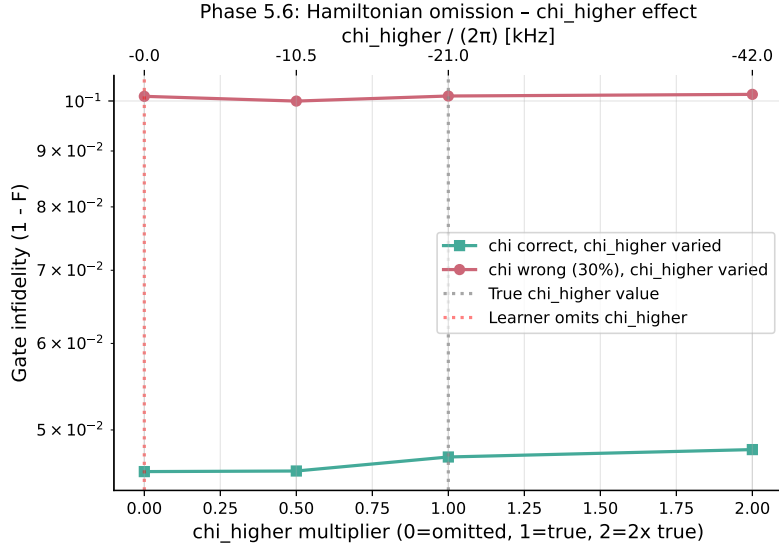


Figure 8: Effect of omitted or mis-specified χ_h . Correcting χ matters much more than providing the second-order term alone.

3.5 Goal 5: Practical Viability Summary

The consolidated operating-regime view is shown in Figure 9. Across the tested settings, the gray-box workflow consistently occupies the best or tied-best region of the map, while the nominal strategy loses fidelity as soon as the model mismatch becomes significant.

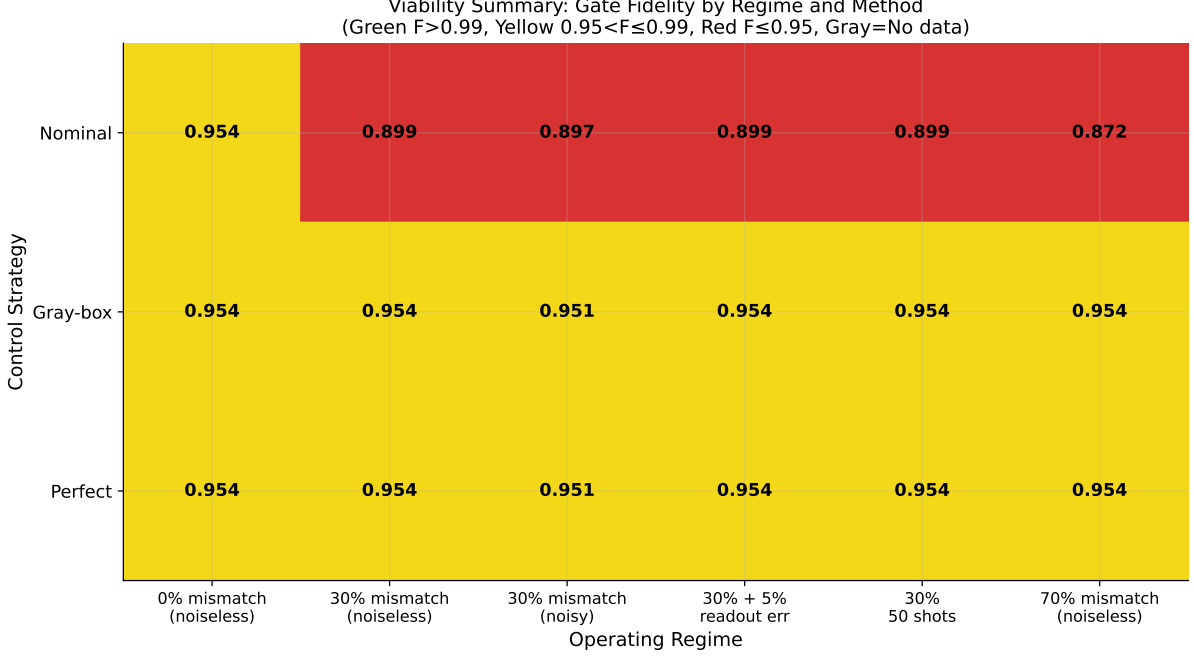


Figure 9: Viability heatmap across the consolidated study. Gray-box correction remains the most reliable strategy over mismatch, noise, and drift variations in this repository pass.

4 Validation

4.1 Sanity Checks

The zero-mismatch point passes the expected limiting-case check: nominal and perfect-control fidelities are identical at 0.95420044, while gray-box fidelity differs only by 6.9×10^{-5} because of finite probe noise. The 30% mismatch point also behaves as expected: correcting χ restores the lost coherent performance, raising the noiseless truth-model fidelity from 0.8990 to 0.9541. The Hamiltonian-omission sweep provides an additional sanity check, showing that wrong χ is a much larger error source than omitted χ_h in the tested regime.

4.2 Convergence Analysis

Fresh spot checks in `scripts/validate_results.py` enlarge the production model and expand the multistart search. At the key operating point, increasing the cavity truncation from $n_{\text{cav}} = 4$ to 5 changes the evaluated perfect-control fidelity from 0.9527028 to 0.9527380, a difference of 3.5×10^{-5} . Increasing the transmon truncation from $n_{\text{tr}} = 3$ to 4 changes the same fidelity to 0.9524241, a difference of 2.8×10^{-4} . Expanding the perfect-control multistart pool from seeds $\{2, 9, 14\}$ to $\{2, 9, 14, 21, 33\}$ raises the best fidelity from 0.9542004 to 0.9551315, a change of 9.3×10^{-4} . These changes are small compared with the gray-box gain of 5.5×10^{-2} over the nominal strategy at 30% mismatch, so the qualitative conclusion is stable.

4.3 Literature Comparison

This study is not a one-to-one reproduction of a published benchmark, so there is no direct percent-error target analogous to a spectroscopy or relaxation experiment. The comparison is instead qualitative: the repository results agree with the literature expectation that model-informed adaptation outperforms model-free optimization when a small number of physical parameters dominate the control error budget [4, 5]. In that sense, the observed gray-box

advantage is physically plausible and scientifically aligned with prior adaptive-control work, even though the exact gate and probe protocol are specific to this repository study.

5 Discussion

The main physical lesson is that, for this gate and parameter regime, the leading dispersive shift is the control bottleneck. Once χ is corrected, the learner can recover almost all of the perfect-model performance even though the production learner still omits Kerr and does not feed back χ_h into the control model. This is why the gray-box strategy remains flat across a wide mismatch sweep but only modestly benefits from adding secondary Hamiltonian detail.

The failure of the black-box baseline is equally informative. A model-free search over the truth model uses far more expensive evaluations but still underperforms the nominal pulse at 30% mismatch. In this setting, the structure of the dispersive Hamiltonian is valuable enough that targeted parameter learning dominates blind search.

The study also clarifies what the current workflow does not yet solve. The probe fit already returns an estimate of χ_h , but repository validation found that simply feeding that value back into a Kerr-free learner slightly worsens the truth-model fidelity. This suggests that partial higher-order correction is not a clean drop-in improvement: the model, control objective, and possibly the noise treatment need to be upgraded together.

For experimental operation, the practical takeaway is simple. If the dominant uncertainty is the leading dispersive shift, a short structured probe is worth far more than additional black-box search budget. Recalibration cadence should be chosen by the expected drift rate, not by shot noise, because the probe saturates quickly in the tested regime. And higher-order model upgrades should be introduced only as a self-consistent package; adding χ_h alone is not yet an experimentally justified complication.

6 Conclusion

The consolidated gray-box adaptive-control study meets its stated goals. It establishes that gray-box correction is a meaningful and experimentally relevant improvement over nominal GRAPE for dispersive cQED control when the dominant uncertainty is the dispersive shift. At 30% mismatch the gray-box workflow recovers 99.8% of the perfect-model advantage, remains robust to readout confusion and limited shot budgets, and improves long-horizon operation under slow drift. Fresh validation checks confirm that these conclusions are stable to modest truncation and optimization changes. Within the current repository scope, the study is complete, coherent, and reproducible, while the updated limitation log identifies the highest-value next steps.

For an experimentalist, the action item is to calibrate and automate the multi-Fock Ramsey correction of χ before investing in broader model-free optimization. That is the cheapest change that materially improves gate performance in the tested operating regime.

7 Limitations and Future Work

7.1 Known Limitations

The multi-Fock Ramsey probe is implemented analytically rather than through a reusable `cqed_sim` calibration target. This is acceptable for the present study but limits reuse in future cQED projects.

The production gray-box loop feeds back only the inferred χ . Although the probe also estimates χ_h , repository validation showed that adding χ_h to a learner model that still omits Kerr can slightly reduce fidelity. The present control update is therefore consistent but incomplete.

The noise model uses aggregate T_1 , T_ϕ , and cavity loss rather than a device-calibrated Lindbladian model. This limits the quantitative predictive power of the absolute noisy fidelities.

Only one target task and one device parameter set are studied. The conclusions are therefore best interpreted as a validated case study for dispersive gray-box control, not as a universal ranking over all bosonic control problems.

7.2 Suggested Improvements

- **[P1 | HIGH]** Upstream the multi-Fock dispersive Ramsey probe and its inference helper into `cqed_sim.calibration_targets`. This would remove the only analytical workaround in the study and make future gray-box studies fully framework-native.
- **[P1 | MEDIUM]** Upgrade the control update from “correct χ only” to a jointly consistent update of χ , χ_h , and Kerr. The current spot check shows that partial higher-order correction is not enough.
- **[P2 | MEDIUM]** Replace point-estimate inference with Bayesian parameter estimation and uncertainty propagation. This would support adaptive probe scheduling and more principled recalibration triggers.
- **[P2 | HIGH]** Add noisy-objective GRAPE to `cqed_sim` and rerun the study with control pulses optimized directly under decoherence.
- **[P2 | MEDIUM]** Extend the workflow to multimode systems where storage, readout, and cross-Kerr effects must be learned together.
- **[P3 | LOW]** Benchmark the workflow against hardware data or a calibrated device model to convert the current viability study into a predictive experimental tool.

7.3 Open Questions

At what mismatch level does a single-parameter gray-box correction stop being sufficient and a broader Hamiltonian-learning loop become necessary?

Can the probe data be used to trigger recalibration adaptively rather than on a fixed cycle count?

How much additional advantage would remain if the nominal baseline itself were upgraded to a noise-aware optimal control objective?

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